Ch02_5

Optimization

Minimization and Karnaugh Maps

- You can simplify, reduce the cost of, the circuit (i.e. logical expression) using two ways:

1. Boolean Algebra properties
2. K-map

- A Karnaugh Map (K-Map) is a visual method used in digital system design to simplify Boolean expressions, reducing hardware complexity and cost.

- It organizes truth table values into a grid.

- A grid representation of the truth table, where adjacent cells differ by only one variable.

- Simplify (or minimize) SOP functions (circuits):

1- Two Variables Map:

$$f(x_1, x_2) = \bar{x}_1\bar{x}_2 + \bar{x}_1 x_2 + x_1 x_2$$

x_1	x_2		f
0	0	m_0	1
0	1	m_1	1
1	0	m_2	0
1	1	m_3	1

x_2 \ x_1	0	1
0	m_0	m_2
1	m_1	m_3

x_1 \ x_2	0	1
0	m_0	m_1
1	m_2	m_3

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2- Three Variables Map:

x_1	x_2	x_3	
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

$x_3 \backslash x_1x_2$	00	01	11	10
0	m_0	m_2	m_6	m_4
1	m_1	m_3	m_7	m_5

$x_1 \backslash x_2x_3$	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Example 1: Simplify the following using a K-map

$$f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + x_1\bar{x}_2x_3 + x_1\bar{x}_3$$

x_1	x_2	x_3	
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

	00	01	11	10
0				
1				

Example 2: Give the minimum cost representation of the following logical expression

$$f(a,b,c) = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b} + ab\bar{c}$$

x_1	x_2	x_3	
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

	00	01	11	10
0				
1				

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Multiple examples:

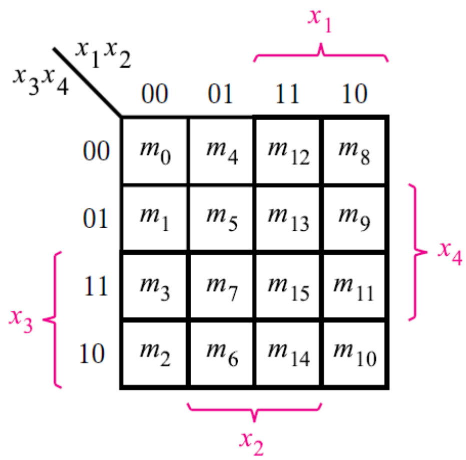
	00	01	11	10
0				
1				

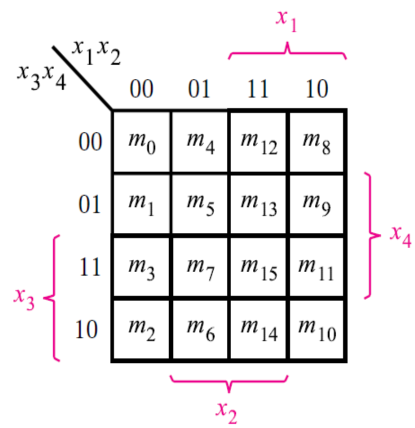
	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

3- Four Variables Map:

X1	X2	X3	X4	f
0	0	0	0	m_0
				m_1
				m_2
				m_3
				m_4
				m_5
				m_6
				m_7
				m_8
				m_9
				m_{10}
				m_{11}
				m_{12}
				m_{13}
				m_{14}
				m_{15}





Example 1: Find the minimum cost SOP expression for the following function. (or Design the simplest SOP expression that implements the following expression)